

19.3 Non-normal errors

Normality entered Part III at one point only: it turned the moments of $\hat{\boldsymbol{\beta}}$ and s^2 into exact distributions (Theorem 7.5.1). Everything that needs only two moments holds for any error distribution with finite variance. Inference about the mean structure survives in large samples, provided no single observation dominates the fit. Inference about σ^2 does not survive at any sample size, and neither does prediction of a single new response.

Throughout, the errors $\varepsilon_1, \dots, \varepsilon_n$ are independent and identically distributed with mean 0 and variance $\sigma^2 \in (0, \infty)$, from a distribution that does not change with n . Where fourth moments are needed, the *excess kurtosis* is $\kappa = \mu_4/\sigma^4 - 3$, where $\mu_4 = E(\varepsilon_i^4)$. It is zero for normal errors, 6 for centred exponential errors, -1.2 for uniform errors, and at least -2 for every distribution, since $\mu_4 \geq \sigma^4$ by Jensen's inequality.

19.3.1 Asymptotic normality of least squares

Large-sample statements concern a sequence of models $\mathbf{Y}_n = \mathbf{X}_n \boldsymbol{\beta} + \boldsymbol{\varepsilon}_n$, $n = 1, 2, \dots$, with p fixed and $\mathbf{X}_n$ of rank p . We write $h_n^* = \max_i h_{ii}$ for the largest leverage of $\mathbf{X}_n$. Besides the three limit theorems recalled in Section 14.4 (the central limit theorem, Slutsky's lemma and the delta method), we need two more standard results of probability theory.

Remark (Two more limit theorems).

- (i) *Lindeberg–Feller*. For each n let $Z_{n1}, \dots, Z_{nn}$ be independent with mean zero and $\sum_i \text{Var}(Z_{ni}) = 1$. If $\sum_i E[Z_{ni}^2 \mathbf{1}\{|Z_{ni}| > \eta\}] \rightarrow 0$ for every $\eta > 0$, then $\sum_i Z_{ni} \rightarrow N(0, 1)$ in distribution (Billingsley 1995, section 27).
- (ii) *Cramér–Wold and continuous mapping*. Random vectors $\mathbf{T}_n$ in $\mathbb{R}^q$ converge in distribution to $\mathbf{T}$ iff $\mathbf{c}^\top \mathbf{T}_n \rightarrow \mathbf{c}^\top \mathbf{T}$ for every $\mathbf{c} \in \mathbb{R}^q$, and then $g(\mathbf{T}_n) \rightarrow g(\mathbf{T})$ for every continuous g (van der Vaart 1998, chapter 2).

Lemma 19.3.1 — Weighted sums of errors

Let $\mathbf{w}_n = (w_{n1}, \dots, w_{nn})^\top$ be constant vectors with $\|\mathbf{w}_n\| = 1$ and $m_n = \max_i |w_{ni}| \rightarrow 0$. Then $\sigma^{-1} \sum_i w_{ni} \varepsilon_i \rightarrow N(0, 1)$ in distribution.

Proof. Put $Z_{ni} = w_{ni} \varepsilon_i / \sigma$, so that $\sum_i \text{Var}(Z_{ni}) = \sum_i w_{ni}^2 = 1$. With $U = \varepsilon_1 / \sigma$, for every $\eta > 0$,

$$\begin{aligned} \sum_i E[Z_{ni}^2 \mathbf{1}\{|Z_{ni}| > \eta\}] &= \sum_i w_{ni}^2 E[U^2 \mathbf{1}\{|w_{ni} U| > \eta\}] \\ &\leq E[U^2 \mathbf{1}\{|U| > \eta/m_n\}], \end{aligned}$$

because $|w_{ni}| \leq m_n$ and the weights w_{ni}^2 sum to one. Since $EU^2 = 1$ and $\eta/m_n \rightarrow \infty$, the right side tends to zero by dominated convergence. The Lindeberg–Feller theorem applies. ■

Theorem 19.3.2 — Least squares with non-normal errors

In the sequence of models above, suppose $h_n^* \rightarrow 0$. Let $s^2 = \|(\mathbf{I} - \mathbf{M})\mathbf{Y}_n\|^2/(n - p)$.

- (a) For every sequence of nonzero vectors $\mathbf{a}_n \in \mathbb{R}^p$,

$$\frac{\mathbf{a}_n^\top \hat{\boldsymbol{\beta}} - \mathbf{a}_n^\top \boldsymbol{\beta}}{\sigma \sqrt{\mathbf{a}_n^\top (\mathbf{X}_n^\top \mathbf{X}_n)^{-1} \mathbf{a}_n}} \rightarrow N(0, 1) \quad \text{in distribution.}$$

- (b) $s^2 \rightarrow \sigma^2$ in probability. This part does not need $h_n^* \rightarrow 0$.
 (c) The t statistic, with s in place of σ in (a), tends to $N(0, 1)$, and the nominal $1 - \alpha$ interval $\mathbf{a}_n^\top \hat{\boldsymbol{\beta}} \pm t_{n-p, \alpha/2} s \sqrt{\mathbf{a}_n^\top (\mathbf{X}_n^\top \mathbf{X}_n)^{-1} \mathbf{a}_n}$ has coverage tending to $1 - \alpha$.
 (d) Let $\boldsymbol{\Lambda}$ be a fixed $p \times q$ matrix of rank q , and suppose $\boldsymbol{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$. The statistic

$$F_n = \frac{(\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})^\top [\boldsymbol{\Lambda}^\top (\mathbf{X}_n^\top \mathbf{X}_n)^{-1} \boldsymbol{\Lambda}]^{-1} (\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})}{q s^2}$$

satisfies $qF_n \rightarrow \chi^2(q)$ in distribution, and $\Pr\{F_n > F_\alpha(q, n - p)\} \rightarrow \alpha$.

Proof.

- (a) Put $\mathbf{G}_n = (\mathbf{X}_n^\top \mathbf{X}_n)^{-1}$ and $\tau_n^2 = \mathbf{a}_n^\top \mathbf{G}_n \mathbf{a}_n$. Since $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = \mathbf{G}_n \mathbf{X}_n^\top \boldsymbol{\varepsilon}_n$, the numerator is $\sum_i c_{ni} \varepsilon_i$ with $c_{ni} = \mathbf{a}_n^\top \mathbf{G}_n \mathbf{x}_{(i)}$, and $\sum_i c_{ni}^2 = \mathbf{a}_n^\top \mathbf{G}_n \mathbf{X}_n^\top \mathbf{X}_n \mathbf{G}_n \mathbf{a}_n = \tau_n^2$. The Cauchy–Schwarz inequality in the inner product $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^\top \mathbf{G}_n \mathbf{v}$ (Proposition 1.1.1 applied to $\mathbf{G}_n^{1/2} \mathbf{a}_n$ and $\mathbf{G}_n^{1/2} \mathbf{x}_{(i)}$) gives

$$c_{ni}^2 \leq (\mathbf{a}_n^\top \mathbf{G}_n \mathbf{a}_n)(\mathbf{x}_{(i)}^\top \mathbf{G}_n \mathbf{x}_{(i)}) = \tau_n^2 h_{ii}.$$

So $w_{ni} = c_{ni}/\tau_n$ has unit length and $\max_i |w_{ni}| \leq \sqrt{h_{ii}^*} \rightarrow 0$. Apply Lemma 19.3.1.

- (b) $(n - p)s^2 = \boldsymbol{\varepsilon}_n^\top \boldsymbol{\varepsilon}_n - \boldsymbol{\varepsilon}_n^\top \mathbf{M} \boldsymbol{\varepsilon}_n$, since $(\mathbf{I} - \mathbf{M})\mathbf{X}_n \boldsymbol{\beta} = \mathbf{0}$. By the weak law of large numbers, $\boldsymbol{\varepsilon}_n^\top \boldsymbol{\varepsilon}_n/n \rightarrow \sigma^2$ in probability. The second term is nonnegative with $E(\boldsymbol{\varepsilon}_n^\top \mathbf{M} \boldsymbol{\varepsilon}_n) = \sigma^2 \text{tr } \mathbf{M} = \sigma^2 p$ (Theorem 2.3.1), so by Markov's inequality $\Pr(\boldsymbol{\varepsilon}_n^\top \mathbf{M} \boldsymbol{\varepsilon}_n/n > \delta) \leq \sigma^2 p/(n\delta) \rightarrow 0$. Since $n/(n - p) \rightarrow 1$, $s^2 \rightarrow \sigma^2$. Under a finite fourth moment this is Corollary 5.6.4; the argument here needs only a finite variance.
 (c) The t statistic is the ratio in (a) multiplied by $\sigma/s \rightarrow 1$, so Slutsky's lemma gives the limit. The quantile $t_{n-p, \alpha/2}$ tends to $z_{\alpha/2}$, because the $t(\nu)$ distribution tends to $N(0, 1)$ and the normal distribution function is continuous and strictly increasing. Hence $|T_n| - t_{n-p, \alpha/2} \rightarrow |Z| - z_{\alpha/2}$, and the coverage tends to $\Pr(|Z| \leq z_{\alpha/2}) = 1 - \alpha$.
 (d) Let $\mathbf{W}_n = \boldsymbol{\Lambda}^\top \mathbf{G}_n \boldsymbol{\Lambda}$, which is positive definite, and $\mathbf{Z}_n = \mathbf{W}_n^{-1/2} \boldsymbol{\Lambda}^\top (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})/\sigma$. Under the hypothesis the numerator of F_n is $\sigma^2 \|\mathbf{Z}_n\|^2$. For a unit vector $\mathbf{c} \in \mathbb{R}^q$, $\mathbf{c}^\top \mathbf{Z}_n = \mathbf{a}_n^\top (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})/\sigma$ with $\mathbf{a}_n = \boldsymbol{\Lambda} \mathbf{W}_n^{-1/2} \mathbf{c}$, and $\mathbf{a}_n^\top \mathbf{G}_n \mathbf{a}_n = \mathbf{c}^\top \mathbf{W}_n^{-1/2} \mathbf{W}_n \mathbf{W}_n^{-1/2} \mathbf{c} = 1$. By (a), $\mathbf{c}^\top \mathbf{Z}_n \rightarrow N(0, 1)$, which is the law of $\mathbf{c}^\top \mathbf{Z}$ for $\mathbf{Z} \sim N_q(\mathbf{0}, \mathbf{I})$. By Cramér–Wold, $\mathbf{Z}_n \rightarrow \mathbf{Z}$, and by continuous mapping $\|\mathbf{Z}_n\|^2 \rightarrow \chi^2(q)$. Then $qF_n = \|\mathbf{Z}_n\|^2 \sigma^2/s^2 \rightarrow \chi^2(q)$ by (b) and Slutsky's lemma. Finally, $qF_\alpha(q, \nu)$ is $\chi^2(q)$ divided by an independent $\chi^2(\nu)/\nu$, which tends to one, so $qF_\alpha(q, n - p)$ tends to the upper α point of $\chi^2(q)$, and the rejection probability tends to α as in (c).

Part (a) allows $\mathbf{a}_n$ to depend on the design. The condition $h_n^* \rightarrow 0$ says that no observation keeps a non-negligible share of the fit. For a straight line at $x = 1, \dots, n$ the largest leverage is below $4/n$ ([Exercise \(A1\)](#)). The condition is not a technicality, as the next example shows.

Example. (An observation with its own parameter)

Fit an intercept together with an indicator of the first observation, $\mathbf{X}_n = [\mathbf{1}, \mathbf{e}_1]$, so that $h_{11} = 1$ for every n . The least squares estimates are $\hat{\mu} = \bar{Y}_{(1)}$, the mean of observations $2, \dots, n$, and $\hat{\delta} = Y_1 - \bar{Y}_{(1)}$. Hence

$$\hat{\delta} - \delta = \varepsilon_1 - \bar{\varepsilon}_{(1)}, \quad \text{Var}(\hat{\delta}) = \sigma^2 \left(1 + \frac{1}{n-1}\right),$$

and since $\bar{\varepsilon}_{(1)} \rightarrow 0$ in probability, $\hat{\delta} - \delta \rightarrow \varepsilon_1$ in distribution. The limit is the error distribution itself. No amount of other data can average away the error of one observation.

19.3.2 How far from normal in a finite sample

[Theorem 19.3.2](#) is a limit statement. A finite-sample measure of the approach comes from the third and fourth cumulants of the numerator, which leverage controls directly.

Proposition 19.3.3 — Skewness and kurtosis of a linear estimate

Let the errors have finite fourth moment, skewness $\gamma_1 = \mu_3/\sigma^3$ with $\mu_3 = E(\varepsilon_i^3)$, and excess kurtosis κ . For $\mathbf{a} \neq \mathbf{0}$ let $Z = (\mathbf{a}^\top \hat{\boldsymbol{\beta}} - \mathbf{a}^\top \boldsymbol{\beta})/\sqrt{\text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}})}$, and let $w_i = c_i/\|\mathbf{c}\|$ with $\mathbf{c} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{a}$. Then

$$E(Z^3) = \gamma_1 \sum_i w_i^3, \quad E(Z^4) - 3 = \kappa \sum_i w_i^4,$$

and $|\sum_i w_i^3| \leq \sqrt{h^*}$, $\sum_i w_i^4 \leq h^*$, where $h^* = \max_i h_{ii}$.

Proof. $Z = \sigma^{-1} \sum_i w_i \varepsilon_i$ with $\sum_i w_i^2 = 1$. Expanding the cube, every term with some index appearing exactly once has mean zero, by independence and $E \varepsilon_i = 0$. Only the terms $w_i^3 \varepsilon_i^3$ remain, so $E(Z^3) = \sigma^{-3} \mu_3 \sum_i w_i^3$. In the fourth power the surviving terms are $w_i^4 \varepsilon_i^4$ and, for $i \neq j$, the 3 pairings of $w_i^2 \varepsilon_i^2 w_j^2 \varepsilon_j^2$. So

$$\begin{aligned} E(Z^4) &= \frac{\mu_4}{\sigma^4} \sum_i w_i^4 + 3 \sum_{i \neq j} w_i^2 w_j^2 \\ &= (\kappa + 3) \sum_i w_i^4 + 3 \left(1 - \sum_i w_i^4\right), \end{aligned}$$

using $\sum_{i \neq j} w_i^2 w_j^2 = (\sum_i w_i^2)^2 - \sum_i w_i^4$. As in the proof of [Theorem 19.3.2\(a\)](#), $w_i^2 \leq h_{ii} \leq h^*$. Hence $|\sum_i w_i^3| \leq \max_i |w_i| \sum_i w_i^2 \leq \sqrt{h^*}$ and $\sum_i w_i^4 \leq \max_i w_i^2 \leq h^*$.

So skewness is averaged away more slowly than heavy tails, although with symmetric weights it can cancel exactly ([Exercise \(B1\)](#)). The denominator s of the t statistic adds terms of the same order, so the proposition describes the leading effect only.

Example. (Coverage of the t interval)

Fit a straight line with normal, centred exponential and scaled $t(3)$ errors, each with variance one, in two designs. In the *spread* design $x = 1, \dots, n$, and h_n^* is 0.345, 0.096 and 0.025 at $n = 10, 40$ and 160 . In the *far-point* design the last value is moved to $x = 4n$, and h_n^* is 0.954, 0.800 and 0.485. With 100000 data sets per case, the coverages of the nominal 95% interval for the slope are:

Design	Errors	$n = 10$	$n = 40$	$n = 160$
spread	normal	0.950	0.951	0.950
spread	exponential	0.955	0.953	0.950
spread	$t(3)$	0.955	0.952	0.951
far point	normal	0.951	0.950	0.951
far point	exponential	0.933	0.948	0.950
far point	$t(3)$	0.931	0.944	0.953

The two-sided coverages hide unbalanced tails. With exponential errors and the far point, the interval misses above the slope in 0.067 of the data sets at $n = 10$ and below it in only 0.001. At $n = 160$ the two proportions are 0.038 and 0.012, still far from 0.025 each, as the slowly falling leverage predicts. A one-sided test in this design has nearly three times its nominal size at $n = 10$. Panel (a) of [Figure 19.3.1](#) shows the skewed t statistic at $n = 40$. The $t(3)$ errors have no fourth moment, but [Theorem 19.3.2](#) still applies.

```
import numpy as np
from scipy import stats

def errors(law, size, rng):
    """Errors with mean 0 and variance 1."""
    if law == "normal":
        return rng.normal(size=size)
    if law == "exponential":
        # skewed: skewness 2, excess
        # kurtosis 6
        return rng.exponential(size=size) - 1.0
    if law == "t3":
        # heavy tails: variance 3 before
        # scaling
        return rng.standard_t(3, size=size) / np.sqrt(3.0)
    if law == "uniform":
        # short tails: excess kurtosis
        # -1.2
        return rng.uniform(-np.sqrt(3), np.sqrt(3), size=size)

def design(kind, n):
    x = np.arange(1.0, n + 1)
    if kind == "far point":
        x[-1] = 4.0 * n
    # one point far to the right
    return np.column_stack([np.ones(n), x])
```

[Run this code online](#)

```
def slope_coverage(kind, law, n, reps, rng):
    """Proportion of nominal 95% t intervals for the slope that cover it."""
    X = design(kind, n)
    G = np.linalg.inv(X.T @ X)
    A = G @ X.T
    E = errors(law, (reps, n), rng)
    B = E @ A.T # beta_hat - beta
    s2 = np.sum((E - B @ X.T) ** 2, axis=1) / (n - 2)
    T = B[:, 1] / np.sqrt(s2 * G[1, 1])
    tq = stats.t.ppf(0.975, n - 2)
    return np.mean(np.abs(T) <= tq), np.mean(T > tq), np.mean(T < -tq)
```

[Run this code online](#)

```
rng = np.random.default_rng(1903)
for kind in ["spread", "far point"]:
    for law in ["normal", "exponential", "t3"]:
        cov, up, lo = slope_coverage(kind, law, 10, 20_000, rng)
        print(f"{kind:9s} {law:11s} n=10: coverage {cov:.3f} (misses above {up:.3f},
              below {lo:.3f})")
```

[Run this code online](#)

19.3.3 Inference about the error variance

The interval for σ^2 (Proposition 12.1.4) rests on SSE, a sum of *squared* errors, whose variance depends on the kurtosis (Proposition 5.6.3, Proposition 7.5.4). A central limit theorem applies, but with the wrong variance.

Proposition 19.3.4 — The interval for sigma squared

In the sequence of models above, let the errors have finite fourth moment with $\kappa > -2$. Then $\sqrt{n}(s^2/\sigma^2 - 1) \rightarrow N(0, 2 + \kappa)$, and the coverage of the nominal $1 - \alpha$ interval of Proposition 12.1.4 tends to

$$2\Phi\left(z_{\alpha/2}\sqrt{2/(2+\kappa)}\right) - 1, \quad (19.3.1)$$

where Φ is the standard normal distribution function. The limit equals $1 - \alpha$ iff $\kappa = 0$.

Proof. As in the proof of [Theorem 19.3.2\(b\)](#),

$$\begin{aligned} \sqrt{n}(s^2 - \sigma^2) &= \frac{n}{n-p}\sqrt{n}\left(\frac{\boldsymbol{\varepsilon}_n^\top \boldsymbol{\varepsilon}_n}{n} - \sigma^2\right) \\ &\quad + \frac{\sqrt{n}p\sigma^2}{n-p} - \frac{\sqrt{n}}{n-p}\boldsymbol{\varepsilon}_n^\top \mathbf{M}\boldsymbol{\varepsilon}_n. \end{aligned}$$

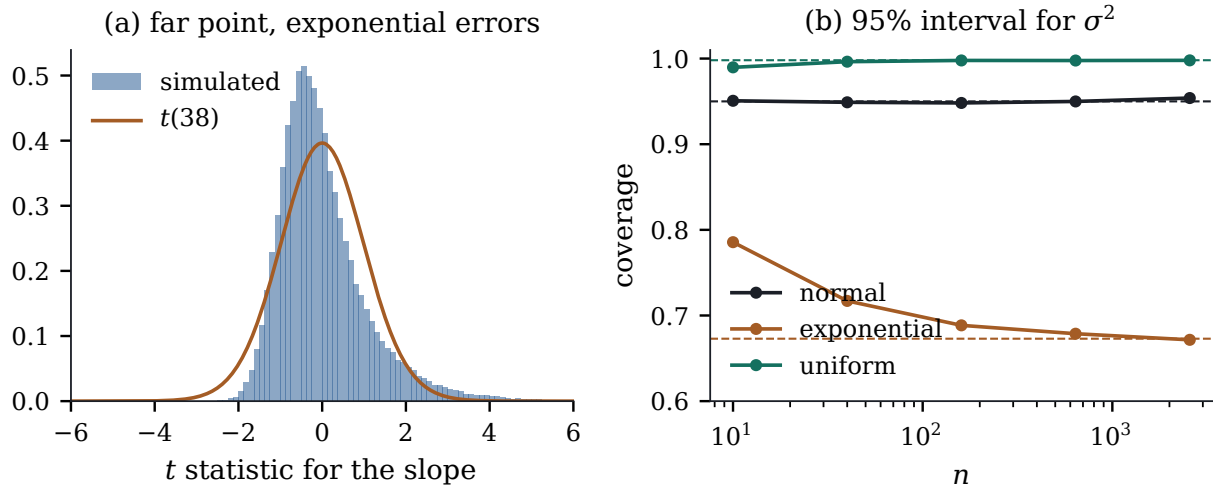


Figure 19.3.1: (a) The t statistic for the slope in the far-point design with $n = 40$ and centred exponential errors (100000 data sets), with the $t(38)$ density. (b) Coverage of the nominal 95% interval for σ^2 in the spread design, against n , with the limits Eq. 19.3.1 dashed.

The ε_i^2 are independent with mean σ^2 and variance $\mu_4 - \sigma^4 = \sigma^4(2 + \kappa)$, so by the central limit theorem the first term tends to $N(0, \sigma^4(2 + \kappa))$. The second term tends to zero, and so does the third in probability, because $E(\mathbf{\varepsilon}_n^\top \mathbf{M} \mathbf{\varepsilon}_n) = \sigma^2 p$ and Markov's inequality applies to $\mathbf{\varepsilon}_n^\top \mathbf{M} \mathbf{\varepsilon}_n / \sqrt{n}$. Slutsky's lemma gives the first claim.

Write $\nu = n - p$ and let c_ν^- and c_ν^+ be the lower and upper $\alpha/2$ points of $\chi^2(\nu)$. The interval covers σ^2 iff $c_\nu^- \leq \nu s^2 / \sigma^2 \leq c_\nu^+$, that is, iff

$$\frac{c_\nu^- - \nu}{\sqrt{\nu}} \leq \sqrt{\nu} \left(\frac{s^2}{\sigma^2} - 1 \right) \leq \frac{c_\nu^+ - \nu}{\sqrt{\nu}}.$$

A $\chi^2(\nu)$ variable is a sum of ν independent squared standard normals, each with variance 2, so by the central limit theorem $(c_\nu^\pm - \nu) / \sqrt{2\nu} \rightarrow \pm z_{\alpha/2}$, and the bounds tend to $\pm z_{\alpha/2} \sqrt{2}$. The middle term tends to $N(0, 2 + \kappa)$ because $\nu/n \rightarrow 1$. The coverage therefore tends to $\Pr(|Z| \sqrt{2 + \kappa} \leq z_{\alpha/2} \sqrt{2})$, which is Eq. 19.3.1.

■

For centred exponential errors, $\kappa = 6$, the nominal 95% interval has limiting coverage $2\Phi(1.96/2) - 1 = 0.673$. For uniform errors, $\kappa = -1.2$, it tends to 0.998. Panel (b) of Figure 19.3.1 shows simulated coverages with the spread design. For exponential errors they are 0.786 at $n = 10$, 0.717 at $n = 40$ and 0.672 at $n = 2560$: more data make it *worse*. This answers Exercise (B3) and explains the failure reported in Section 12.1.

```
def sigma2_coverage(law, n, reps, rng):
    X = design("spread", n)
    E = errors(law, (reps, n), rng)
    B = np.linalg.solve(X.T @ X, X.T @ E.T).T
    sse = np.sum((E - B @ X.T) ** 2, axis=1) # sigma^2 = 1
    lo, hi = stats.chi2.ppf([0.025, 0.975], n - 2)
    return np.mean((sse >= lo) & (sse <= hi))

def limit(kappa, z=stats.norm.ppf(0.975)):
    """Limiting coverage of the nominal 95% chi-squared interval."""
    return 2 * stats.norm.cdf(z * np.sqrt(2 / (2 + kappa))) - 1
```

```
print("limits:", {k: round(limit(k), 3) for k in [0, 6, -1.2]})
```

[Run this code online](#)

Prediction intervals fail for the same reason: they must cover a single new error, which no averaging acts on (Proposition 12.3.3).

19.3.4 Why the F test is protected and the variance-ratio test is not

The F statistic is a ratio of two quadratic forms, each with a kurtosis-dependent variance. The effects can cancel, as Box and Watson (1962) and Atiqullah (1962) found. Since $\log F$ is approximately the difference of the two mean squares divided by σ^2 , it suffices to study that difference.

Proposition 19.3.5 — Kurtosis and a ratio of mean squares

Let $\mathbf{P}_1$ and $\mathbf{P}_2$ be symmetric idempotent with $\mathbf{P}_1\mathbf{P}_2 = \mathbf{0}$, ranks f_1 and f_2 , and $\mathbf{P}_1\mathbf{X}\boldsymbol{\beta} = \mathbf{P}_2\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$. Let the errors have finite fourth moment and excess kurtosis κ , put $S_j^2 = \mathbf{Y}^\top \mathbf{P}_j \mathbf{Y} / f_j$, and let $\mathbf{p}_j$ be the vector of diagonal entries of $\mathbf{P}_j$. Then $L = (S_1^2 - S_2^2) / \sigma^2$ has mean zero and

$$\text{Var}(L) = \frac{2}{f_1} + \frac{2}{f_2} + \kappa \|\mathbf{d}\|^2, \quad \mathbf{d} = \frac{\mathbf{p}_1}{f_1} - \frac{\mathbf{p}_2}{f_2}.$$

In particular, if both $\mathbf{P}_1$ and $\mathbf{P}_2$ have constant diagonals, then $\mathbf{d} = \mathbf{0}$ and $\text{Var}(L)$ is the same as under normality.

Proof. Since $\mathbf{P}_j \mathbf{Y} = \mathbf{P}_j \boldsymbol{\varepsilon}$, $L = \boldsymbol{\varepsilon}^\top \mathbf{A} \boldsymbol{\varepsilon} / \sigma^2$ with $\mathbf{A} = \mathbf{P}_1 / f_1 - \mathbf{P}_2 / f_2$, whose diagonal is $\mathbf{d}$. By Theorem 2.3.1, $E(L) = \text{tr} \mathbf{A} = 1 - 1 = 0$. By Theorem 2.3.3 with zero means, $\text{Var}(\boldsymbol{\varepsilon}^\top \mathbf{A} \boldsymbol{\varepsilon}) = (\mu_4 - 3\sigma^4) \mathbf{d}^\top \mathbf{d} + 2\sigma^4 \text{tr}(\mathbf{A}^2)$, and $\text{tr}(\mathbf{A}^2) = \text{tr} \mathbf{P}_1 / f_1^2 + \text{tr} \mathbf{P}_2 / f_2^2 - 2\text{tr}(\mathbf{P}_1 \mathbf{P}_2) / (f_1 f_2) = 1/f_1 + 1/f_2$. Divide by σ^4 . If the diagonals are constant, $\mathbf{p}_j = (f_j/n)\mathbf{1}$ because $\text{tr} \mathbf{P}_j = f_j$, and $\mathbf{d} = \mathbf{0}$. ■

For the F test of a linear hypothesis, $\mathbf{P}_1 = \mathbf{M} - \mathbf{M}_0$ and $\mathbf{P}_2 = \mathbf{I} - \mathbf{M}$. Designs in which both have constant diagonals are called *quadratically balanced*. Every balanced one-way or complete factorial layout is an example. In such designs the kurtosis of the errors does not affect the F statistic to first order. The variance-ratio test for comparing the error variances of two independent samples is the opposite case. With n_1 and n_2 observations, each sample centred at its own mean, $\mathbf{d}$ has entries $1/n_1$ on the first sample and $-1/n_2$ on the second (Exercise (B2)), so

$$\text{Var}(L) = \frac{2}{f_1} + \frac{2}{f_2} + \kappa \left(\frac{1}{n_1} + \frac{1}{n_2} \right),$$

and the kurtosis term is as large as the normal-theory term. The test is not robust at any sample size. For exponential errors and $n_1 = n_2 = 20$, the nominal 5% test rejected equal variances in 0.268 of 100000 simulated pairs of samples. Its large-sample limit is 0.327 (Exercise (B2)). The

F test for equal means in a balanced one-way layout with three groups of eight had size 0.042, and with unbalanced groups of 2, 4 and 18 it had size 0.058. In the unbalanced layout $\text{Var}(L)$ is 1.870 instead of 1.095, which overstates the effect on the size: in the tail, skewness and the curvature of the logarithm also matter.

Idea. Inference about β is robust to the *shape* of the errors when no observation has large leverage. Inference about σ^2 , variance ratios or single future responses is not, at any sample size.

19.3.5 Exercises

19.3.5.1 A. Check your understanding

Exercise A1.

For a straight line at $x = 1, \dots, n$, show that the largest leverage is $1/n + 3(n-1)/(n(n+1)) < 4/n$.

Exercise A2.

In the far-point design of [Example \(Coverage of the \$t\$ interval\)](#), show that the leverage of the far point is approximately $147/(n + 147)$ for large n . Why does the table still show nearly nominal two-sided coverage at $n = 160$, when the leverage is almost one half?

19.3.5.2 B. Practice

Exercise B1.

In a one-way layout with m observations per group and skewed errors, compute $\sum_i w_i^3$ of [Proposition 19.3.3](#) for (i) a single group mean and (ii) the difference of two group means. What does this say about one-sample and two-sample t tests with skewed data?

Solution.

- (i) The mean of group 1 has $c_i = 1/m$ on its m observations, so $w_i = m^{-1/2}$ there and $\sum_i w_i^3 = m \cdot m^{-3/2} = m^{-1/2}$. (ii) The difference has $c_i = \pm 1/m$ on the two groups, so $w_i = \pm(2m)^{-1/2}$, and the cubes cancel: $\sum_i w_i^3 = 0$. The numerator of a one-sample t statistic has skewness $\gamma_1/\sqrt{m}$. That of a two-sample comparison with equal group sizes has none, and its non-normality starts with the kurtosis term, of order $1/m$. Equal-sized two-sample comparisons are therefore much better protected against skewness.

Exercise B2.

Two independent samples of size m are each centred at their own mean. Let S_1^2 and S_2^2 be the sample variances. (i) Show that the vector $\mathbf{d}$ of [Proposition 19.3.5](#) has entries $1/m$ and $-1/m$. (ii) Using the first part of [Proposition 19.3.4](#) for each sample, show

that when the two variances are equal the two-sided variance-ratio test of nominal size α has size tending to $2(1 - \Phi(z_{\alpha/2}\sqrt{2/(2+\kappa)}))$ as $m \rightarrow \infty$.

Solution.

- (i) $\mathbf{P}_1$ is the centring matrix on the first sample and zero elsewhere, so its diagonal is $1 - 1/m = f_1/m$ there, and $\mathbf{p}_1/f_1$ has entries $1/m$ on the first sample. Likewise $\mathbf{p}_2/f_2$ has entries $1/m$ on the second, which enter $\mathbf{d}$ with a minus sign. (ii) By [Proposition 19.3.4](#) (with $p = 1$), $\sqrt{m}(S_j^2/\sigma^2 - 1) \rightarrow N(0, 2 + \kappa)$ for $j = 1, 2$, independently. By the delta method, $\sqrt{m} \log(S_1^2/S_2^2) \rightarrow N(0, 2(2 + \kappa))$. Under normality the same statistic tends to $N(0, 4)$, and the critical values of the $F(m - 1, m - 1)$ distribution satisfy $\sqrt{m} \log F_{\alpha/2} \rightarrow 2z_{\alpha/2}$. The rejection probability therefore tends to $\Pr(|Z|\sqrt{2(2 + \kappa)} > 2z_{\alpha/2}) = 2(1 - \Phi(z_{\alpha/2}\sqrt{2/(2 + \kappa)}))$. For $\kappa = 6$ and $\alpha = 0.05$ this is 0.327.

19.3.5.3 C. Going deeper

Exercise C1.

Let the errors be independent with mean zero, variances $\sigma_i^2 \in [m_0, M_0]$ with $m_0 > 0$, and $\sup_i E|\varepsilon_i|^3 < \infty$. Suppose $h_n^* \rightarrow 0$. Show that for every nonzero $\mathbf{a}_n$, $(\mathbf{a}_n^\top \hat{\boldsymbol{\beta}} - \mathbf{a}_n^\top \boldsymbol{\beta})/\sqrt{\text{Var}(\mathbf{a}_n^\top \hat{\boldsymbol{\beta}})} \rightarrow N(0, 1)$, where the variance is the sandwich Eq. 5.5.3. Explain why this does not make the usual t statistic asymptotically standard normal.

Solution. With c_{ni} as in the proof of [Theorem 19.3.2](#), $\text{Var}(\mathbf{a}_n^\top \hat{\boldsymbol{\beta}}) = \sum_i c_{ni}^2 \sigma_i^2 \geq m_0 \tau_n^2$. Put $Z_{ni} = c_{ni} \varepsilon_i / \sqrt{\text{Var}(\mathbf{a}_n^\top \hat{\boldsymbol{\beta}})}$. Lyapunov's bound $E[Z^2 \mathbf{1}\{|Z| > \eta\}] \leq E|Z|^3/\eta$ gives the Lindeberg sum at most

$$\begin{aligned} & \frac{\sup_i E|\varepsilon_i|^3 \sum_i |c_{ni}|^3}{\eta (m_0 \tau_n^2)^{3/2}} \\ & \leq \frac{\sup_i E|\varepsilon_i|^3}{\eta m_0^{3/2}} \max_i \frac{|c_{ni}|}{\tau_n} \\ & \leq \frac{\sup_i E|\varepsilon_i|^3}{\eta m_0^{3/2}} \sqrt{h_n^*} \rightarrow 0, \end{aligned}$$

using $\sum_i |c_{ni}|^3 \leq \max_i |c_{ni}| \tau_n^2$. The usual t statistic divides by $s \tau_n$, and $s^2 \tau_n^2$ estimates $\bar{\sigma}^2 \tau_n^2$ with $\bar{\sigma}^2$ an average variance, not $\sum_i c_{ni}^2 \sigma_i^2$. The ratio of the two is the quantity of Exercise (B1), and it need not tend to one. [Chapter 21](#) replaces $s^2 \tau_n^2$ by a consistent estimate of the sandwich.